

Homework 7

Exercises

1. For $a, b \in \mathbb{Z}^+$, their **least common multiple** is the smallest $m \in \mathbb{Z}^+$ such that $a|m$ and $b|m$. Note that such a number must exist by Well-Ordering; we write $m = \text{lcm}(a, b)$. Prove, for all $a, b, c \in \mathbb{Z}^+$, that $a|c$ and $b|c$ means $\text{lcm}(a, b)|c$.

Hint: Try dividing c by $m = \text{lcm}(a, b)$. Can you prove that the remainder is 0?

2. Let $p, q \in \mathbb{Z}^+$ be prime. Show directly from the definition of *prime* (not irreducible!) that if $p|q$, then $p = q$.
3. Let $a, b \in \mathbb{Z}^+$. If we write $p_1, \dots, p_r$ for the primes that appear in at least one of a and b 's prime factorization, we can say

$$a = p_1^{e_1} \cdots p_r^{e_r} \quad \text{and} \quad b = p_1^{f_1} \cdots p_r^{f_r}$$

for $e_i, f_i \in \mathbb{N}$. With this formulation, their greatest common divisor and least common multiple are given by

$$\text{gcd}(a, b) = p_1^{\min(e_1, f_1)} \cdots p_r^{\min(e_r, f_r)} \quad \text{and} \quad \text{lcm}(a, b) = p_1^{\max(e_1, f_1)} \cdots p_r^{\max(e_r, f_r)}.$$

In particular, we always have the following formula: $\text{gcd}(a, b) \text{lcm}(a, b) = ab$.

Compute $\text{lcm}(1495, 650)$ by using the Euclidean algorithm to find $\text{gcd}(1495, 650)$ and again by using prime factorizations.

4. Prove that, for all $a, b \in \mathbb{Z}^+$, that $a^2|b^2$ means $a|b$.

Hint: There are many approaches here. Try prime factorization!

5. Let $x \in \mathbb{Z}$.

- (a) Prove that if x is even, then $x^2 \equiv 0$ or $4 \pmod{8}$.
- (b) Prove that if x is odd, then $x^2 \equiv 1 \pmod{8}$.

6. Take $n \in \mathbb{Z}^+$ and express it in base ten: $n = (d_k \cdots d_1 d_0)_{10}$, with each $d_i \in \{0, 1, \dots, 9\}$, which means

$$n = 10^k d_k + \cdots + 10d_1 + d_0.$$

- (a) Explain why $3|n$ if and only if $3|(d_k + \cdots + d_1 + d_0)$. Note that this gives a convenient way to check if a number, written out in base 10, is divisible by 3: add up its digits and check if the sum is divisible by 3.

Hint: Can we rephrase this equivalence using modular arithmetic?

- (b) Show that 102, 715, 369, 770, 375 is divisible by 3.

7. Let $p = 37$ and $g = 17$.

- (a) Suppose Alice chooses the secret $a = 4$ and Bob chooses $b = 7$. Compute both of their public keys, $A = g^a \pmod{p}$ and $B = g^b \pmod{p}$, and their shared secret $S = g^{ab} \pmod{p}$ (from both Alice and Bob's perspectives).
- (b) Compute S^{-1} modulo p using the Euclidean algorithm.
- (c) Eve learns Carlos' public key: $C = 5 \pmod{31}$. Use a brute-force attack to determine Carlos' secret key c .

Remark: The point of this problem is to demonstrate how, even for small primes, the discrete logarithm problem requires some work to solve. That said, you are encouraged to use a calculator, or perhaps Python's built-in functions for modular arithmetic, to speed up your search. You can compute remainders modulo p using the modulo operator `%` in Python.

(d) (*Bonus*) Returning to Alice's perspective, let's represent Arabic numerals and English letters via the rule:

$$"0" \mapsto 0, \quad "1" \mapsto 1, \quad \dots, \quad "9" \mapsto 9, \quad "A" \mapsto 10, \quad "B" \mapsto 11, \quad \dots, \quad "Z" \mapsto 35, \quad \text{and} \quad " " \mapsto 36.$$

Bob writes a message using these limited characters, encoding each character into numbers via the above correspondence, then encrypts the message by multiplying each value by S modulo p . He then converts the message back into characters and transmits the result to Alice.

Suppose that Bob sends Alice the message "Q6MI7I2TBJ". Decode the result using S^{-1} .

Note: This “encryption” method (multiplying each letter by S) is highly insecure—we call it a [monoalphabetic substitution cipher](#), which is easily defeated by [frequency analysis](#). In real-world cryptographic schemes, the shared secret from Diffie–Hellman is not used directly as a key, but rather as an input to a [Key Derivation Function](#). Secure encryption schemes also use randomization so that the same letter encrypts differently each time. For more, see your nearest cryptography class!

References